

The Larsen Empirical Hydrogen-Bond Shielding Calculator

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1 What the calculator computes

The Larsen hydrogen-bond calculator evaluates the empirical ProCS15 hydrogen-bond term for each atom in a protein conformation. It does not run an electronic-structure calculation. It finds geometrically plausible hydrogen-bond pairs, looks up precomputed DFT response grids indexed by that geometry, rotates the returned 3×3 tensors into the protein frame, and accumulates the atom-specific terms that Larsen's table assigns to each backbone atom type.

The physical effect being represented is the change in nuclear shielding caused when a donor hydrogen approaches an acceptor oxygen. A hydrogen bond polarizes the local electronic structure of the donor and acceptor groups; those electrons then change the secondary magnetic field at nearby nuclei. The observable connected to this calculation is the shielding tensor σ , usually reported experimentally after conversion to chemical shift. In this implementation the grid tensors and the water offset are in ppm, because the grid values are DFT shielding differences supplied by the Larsen parameterization. The calculator maps structure to tabulated tensors; it does not run the electronic response calculation or convert those tensors into a complete chemical shift.

2 Where shielding comes from

For a nucleus placed in an applied magnetic field B_0 , the local field is written

$$B_{\text{nuc}} = (I - \sigma)B_0. \quad (1)$$

I is the 3×3 identity matrix. The shielding tensor σ is dimensionless in physical NMR and is normally quoted in parts per million. It is a linear-response coefficient: if the applied field is changed a little, σ tells how much of the induced electronic field opposes or redirects that change at the nucleus.

The tensor form is needed because molecules are oriented objects. A field applied along the peptide plane, for example, can induce a different secondary field than a field applied normal to that plane. A matrix records this directional response: the entry σ_{ab} says how an applied-field component along Cartesian direction b contributes to the screened field component along direction a .

At the quantum-mechanical level, Eq. (1) comes from the electrons responding to B_0 . The perturbation changes their current and spin response, and those electrons produce an induced magnetic field at the nucleus. The Larsen calculator keeps a local empirical piece of that response: the portion associated with hydrogen-bond geometry. Its input coordinates are not electronic wave functions, but the hydrogen-bond distance and angles that select a precomputed DFT difference tensor.

Solution NMR often reports an isotropic chemical shift, related to the average shielding after reference subtraction and sign convention. The isotropic number is not the whole tensor. Hydrogen bonding also changes the orientation-dependent part of the shielding, and the code preserves that rank-2 information rather than reducing the grid lookup to a scalar shift.

3 The tensor split used by the code

Every accumulated Larsen tensor is decomposed by the C++ type `SphericalTensor`. For a real 3×3 tensor X , the scalar part is

$$T_0 = \frac{1}{3} \text{tr } X, \quad (2)$$

the average of the three diagonal entries. This is the isotropic part; it is the component that survives rotational tumbling.

The antisymmetric part is stored as a three-component pseudovector,

$$\mathbf{T}_1 = \frac{1}{2} \begin{pmatrix} X_{yz} - X_{zy} \\ X_{zx} - X_{xz} \\ X_{xy} - X_{yx} \end{pmatrix}. \quad (3)$$

It stores the part of the matrix that changes sign under transpose. In routine isotropic solution NMR this term is not usually observed directly, but the code carries it because the grid returns full matrices.

The remaining symmetric traceless matrix is

$$\mathbf{S} = \frac{\mathbf{X} + \mathbf{X}^\top}{2} - T_0 \mathbf{I}, \quad \text{tr } \mathbf{S} = 0. \quad (4)$$

It has five independent entries. The code stores those five entries in the real spherical basis

$$\mathbf{T}_2 = \begin{pmatrix} \sqrt{2} S_{xy} \\ \sqrt{2} S_{yz} \\ \sqrt{3/2} S_{zz} \\ \sqrt{2} S_{xz} \\ (S_{xx} - S_{yy})/\sqrt{2} \end{pmatrix}. \quad (5)$$

The factors in Eq. (5) are the implementation's normalization. In this basis, the squared length of the five-vector equals the Frobenius norm of the matrix $\mathbf{S}$:

$$\sum_m T_{2m}^2 = \sum_{ab} S_{ab}^2. \quad (6)$$

The nine written doubles are ordered as T_0 , the three T_1 components, and the five T_2 components. Larsen-HBond populates all three pieces whenever the grid tensor contains them. The separate water term is scalar in ppm and is written outside this tensor total.

4 The method implemented

The runtime first enumerates donor hydrogens. A backbone amide hydrogen $\text{HN}(i)$ is an amide donor when the bond graph supplies the preceding residue's carbonyl carbon; its donor-frame anchors are $\text{N}(i)$ and $\text{C}'(i-1)$. An alpha hydrogen $\text{H}\alpha(i)$, including both HA2 and HA3 on glycine, is an alpha donor; its anchors are $\text{C}\alpha(i)$ and $\text{N}(i)$. Candidate acceptor oxygen atoms are found within 4.2 Å of the donor hydrogen, excluding atoms in the donor's own residue.

An acceptor oxygen is then classified by its chemistry. Backbone carbonyl acceptors use the NMA acceptor grid. Asn/Gln side-chain amide oxygens are approximated by that same backbone-carbonyl grid, because the Larsen archive has no separate side-chain-amide acceptor scan. Hydroxyl acceptors from Ser, Thr, and Tyr use the methanol grid. Asp/Glu and terminal carboxylate oxygens use the acetate grid, with the closer of the two carboxylate oxygens selected as the accepting oxygen. These choices select one of six archives: NMANMA, NMACOH, NMACOO, ALANMA, ALACOH, and ALACOO.

For a donor hydrogen position $\mathbf{h}$, acceptor oxygen $\mathbf{o}$, acceptor heavy atom $\mathbf{c}$, and acceptor-side third atom $\mathbf{t}$, the lookup coordinates are

$$r = \|\mathbf{o} - \mathbf{h}\|, \quad \theta = \cos^{-1} \left(\frac{(\mathbf{h} - \mathbf{o}) \cdot (\mathbf{c} - \mathbf{o})}{\|\mathbf{h} - \mathbf{o}\| \|\mathbf{c} - \mathbf{o}\|} \right), \quad (7)$$

and

$$\rho = \text{atan2} \left([\mathbf{n}_1 \times (\widehat{\mathbf{c} - \mathbf{o}})] \cdot \mathbf{n}_2, \mathbf{n}_1 \cdot \mathbf{n}_2 \right), \quad \mathbf{n}_1 = (\mathbf{o} - \mathbf{h}) \times (\mathbf{c} - \mathbf{o}), \quad \mathbf{n}_2 = (\mathbf{c} - \mathbf{o}) \times (\mathbf{t} - \mathbf{c}). \quad (8)$$

Here r is in Å, while θ and ρ are in degrees in the code. θ is the angle H–O–C at the acceptor oxygen. ρ is the signed H–O–C–third dihedral, using the IUPAC `atan2` convention. The Larsen archive filenames use the opposite ρ sign, but the parser keyed the dense grids on the IUPAC sign; the runtime therefore does not negate ρ . The code accepts a geometric hydrogen bond when $\theta \in [90^\circ, 180^\circ]$. If θ passes but the lookup is outside the selected grid’s stored r or θ range, the pair is counted as a grid miss; it still suppresses the water term for that amide hydrogen because the geometry was hydrogen-bond-like.

The grid lookup is tensor-valued trilinear interpolation. For the selected donor and acceptor classes, the archive stores tensors $G_{ijk}^{(q)}$ on a regular grid in r, θ, ρ , where q names the readout atom such as donor HN, donor C, or acceptor CA. The runtime locates the enclosing cell and returns

$$G^{(q)}(r, \theta, \rho) = \sum_{\alpha, \beta, \gamma \in \{0,1\}} w_{\alpha\beta\gamma} G_{i+\alpha, j+\beta, k+\gamma}^{(q)} \quad (9)$$

The weights $w_{\alpha\beta\gamma}$ are the usual products of the fractional distances along the three axes and sum to one. The ρ axis wraps periodically at $\pm 180^\circ$; r and θ must lie inside their stored ranges, with a small floating-point tolerance at the bounds. The actual metadata place the NMA-donor r axes near 1.5 Å to 4.0 Å, the ALA-donor axes near 1.6 Å to 4.0 Å or 1.8 Å to 4.0 Å depending on acceptor class, and the θ axes near 90° to 180° . Some dense-grid cells were filled upstream for continuity; the calculator propagates a flag when any of the eight interpolation corners was imputed.

The returned tensors are stored in a donor-centered frame. The frame rotation is built from the donor atoms:

$$\hat{z} = \frac{\mathbf{h} - \mathbf{a}}{\|\mathbf{h} - \mathbf{a}\|}, \quad \hat{x} = \frac{(\mathbf{h} - \mathbf{u}) - [(\mathbf{h} - \mathbf{u}) \cdot \hat{z}] \hat{z}}{\|(\mathbf{h} - \mathbf{u}) - [(\mathbf{h} - \mathbf{u}) \cdot \hat{z}] \hat{z}\|}, \quad \hat{y} = \hat{z} \times \hat{x}. \quad (10)$$

$\mathbf{a}$ is the donor anchor and $\mathbf{u}$ is the donor-side third atom listed above. Eq. (10) is a Gram–Schmidt projection: it makes an orthonormal basis by keeping the H–anchor direction as the z -axis and removing that z -component from the H–third direction before making the x -axis. With R storing $\hat{x}, \hat{y}, \hat{z}$ as its rows, the protein-frame tensor is

$$\sigma_{\text{lab}} = R^T \sigma_{\text{canon}} R. \quad (11)$$

This is the standard rank-2 change of basis: both matrix indices rotate.

The per-atom dispatch then follows the Larsen contribution table encoded in `LarsenContribDispatch`. Amide-H donors contribute primary HB terms to their own residue and secondary HB terms to the NMA acceptor side; alpha-H donors contribute the corresponding $H\alpha B$ terms. The implemented mask is: N receives 1°HB , 2°HB , and $2^\circ H\alpha B$; $C\alpha$ and C' receive 2°HB ; $C\beta$ is excluded from the physical sum; $H\alpha$ and HN receive all four pair terms. Backbone NMA acceptor readouts route N, HN, CA, and HA to residue $j + 1$, while the acceptor C readout routes to the accepting residue j . Hydroxyl and carboxylate grids do not carry acceptor-side readouts. Side-chain carbonyls reuse the NMA grid, but the code has no backbone $j + 1$ residue to route to, so it does not apply secondary terms for those acceptors either.

For atom a , the tensor total written by the calculator is

$$\sigma_a^{\text{pairs}} = \sigma_a^{1^\circ\text{HB}} + \sigma_a^{2^\circ\text{HB}} + \sigma_a^{1^\circ H\alpha B} + \sigma_a^{2^\circ H\alpha B}. \quad (12)$$

Each term is a sum of rotated grid tensors assigned to that atom. The $C\beta$ readout is rotated and written as a diagnostic tensor, but it is not included in Eq. (12) and does not increment the per-atom pair count. Amide hydrogens with no successful grid hit and no post- θ -gate grid miss receive a separate water offset

$$\Delta\sigma_w = 2.07 \text{ ppm}. \quad (13)$$

The code stores this as a scalar field on HN atoms, not as an addition to σ_a^{pairs} .

There are no Larsen-specific numeric keys in `data/calculator_params.toml`. The TOML H-bond entries there (50.0 Å dipolar maximum distance, 3.5 Å counting radius, and a two-residue sequence exclusion) belong to other hydrogen-bond calculators. `LarsenHBond` instead uses the compiled constants 4.2 Å, 90° , 180° , and 2.07 ppm, plus the supplied grid axes.

5 Inputs and output

The runtime inputs are atom positions in Å, residue atom slots, the bond graph, typed atom semantics, and the spatial atom index. It does not consume MOPAC charges, AIMNet2 charges or embeddings, ff14SB charges, APBS fields, or DSSP hydrogen-bond assignments. Those are separate external data streams in the wider pipeline. LarsenResidue perception is also not in this calculator’s runtime path.

The external data used here are the six dense HDF5 grids under `data/larsen_hbond_grids`, or the directory supplied through the runtime `larsen_hbond_grids` path. They are externally supplied tabulated Larsen parameterization data. This document treats them as lookup tables and does not rederive how they were fit.

`WriteFeatures` emits eight files: total pair tensor, the four per-class pair tensors, the $C\beta$ diagnostic tensor, the water term, and the per-atom pair count. The six tensor files are $N \times 9$ tables in the T_0, T_1, T_2 order of Eqs. (2)–(5); the water file is $N \times 1$ in ppm; the count file is a one-dimensional integer array. The tensors are empirical, geometry-indexed shielding-difference tensors from the Larsen grids, not a fresh prediction of the chemical shift. There is no separate fitted scalar applied after lookup inside this calculator; later comparison with DFT reference shieldings belongs to the analysis that uses these tensor features.